LIVING THINGS

Food Chains and Energy Roles

Producers, consumers and decomposers, building food chains and predicting changes



grass **grasshopper** **frog**

In this pack

- **Learn it:** Food chains: producers, consumers and decomposers ★☆☆
- **Fact check:** Who does what? ★☆☆
- **Think it through:** What if…? ★★☆
- **Investigate:** What makes bread go mouldy? ★★☆
- **Real-world case study:** The frogs and the rice field ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Food chains: producers, consumers and decomposers

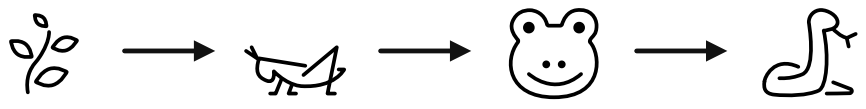
★☆☆ Grade 3

Name: Class: Date:

Learning goals

- Explain the jobs of producers, consumers and decomposers.
- Build food chains from what we observe.
- Predict what happens when one living thing changes.

Every living thing needs food for energy. A **food chain** shows who eats whom. The arrows point from the food to the animal that eats it.



grass *producer* → **grasshopper** *consumer* → **frog** *consumer* → **snake** *consumer*

A food chain in a rice field.

Producers

Producers are green plants. They **make their own food** using **sunlight**, water and air. Every food chain starts with a producer.

Consumers

Consumers are animals. They **eat** living plants or animals. A goat eats grass. A frog eats grasshoppers. A tiger eats deer.

Decomposers: nature's recyclers

When plants and animals die, or leaves fall, **decomposers** break them down. **Moulds**, **mushrooms** (fungi), **worms**, some **insects** and tiny **microbes** are decomposers. They turn dead things back into goodness in the **soil**, which helps new plants grow. Without decomposers, dead leaves and animals would pile up everywhere!



mushrooms



earthworms



flies and beetles



microbes and mould

When something changes

If one living thing in a food chain disappears, the others change too. If farmers kill all the frogs, the grasshoppers have no predator, so there will be **more grasshoppers**, which eat **more rice**. And the snakes will have **less food**.

Remember

Patterns help us predict: when a predator goes, its prey often increases.

Key words

producer	a plant that makes its own food using sunlight
consumer	an animal that eats other living things
decomposer	breaks down dead things into the soil
predator	an animal that hunts others
prey	an animal that is eaten

FACT CHECK · A5 · FOOD CHAINS

Who does what?

★☆☆
Grade 3

Name: Class: Date:

Learning goals

- Sort producers, consumers and decomposers.
- Build food chains.

1 Sort them. [8]

SORT THESE

earthworm tiger mango tree grass mould goat mushroom frog

Producer	Consumer	Decomposer

2 Put these in order to make a food chain. [2]



tiger



grass



deer

3 Fill in the gaps. [3]

WORD BANK

decomposers producers consumers predators

Plants are _____. Animals that eat food are _____. Worms and mushrooms are _____.

THINK IT THROUGH · A5 · FOOD CHAINS

What if...?

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Predict changes in food chains.

1 **In the chain grass → grasshopper → frog → snake, what would happen if all the frogs were gone?** [3]

a) to the grasshoppers?

b) to the grass?

c) to the snakes?

2 **Why are decomposers important?** [2]

3 **Draw a food chain you can see near your school.** [2]

My food chain

INVESTIGATE · A5 · FOOD CHAINS

What makes bread go mouldy?

★★☆ Grade 3

Name: Class: Date:

Learning goals

- Do a fair test.
- Observe changes over time.

Class 3C put three slices of bread in bags: one dry, one damp, one damp in the fridge. They looked every day for a week.

1 **Results: days until mould appeared.** [1]

Bread	dry	damp (warm room)	damp (fridge)
Days	no mould after 7 days	3	7

2 **Which bread went mouldy first? Why?** [2]

.....

.....

3 **Is mould a producer, consumer or decomposer?** [1]

.....

Stay safe

Never open the bags or smell mouldy bread. Throw them away sealed.

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REAL-WORLD CASE STUDY · A5 · FOOD CHAINS

The frogs and the rice field

★★★
Grade 3

Name: **Class:** **Date:**

Learning goals

- Use a food chain to explain a real problem.

Science | Farming | Environment

Frogs live in many rice fields in Bangladesh. They eat insects that damage the rice. Some years ago, many people caught frogs to sell, and many farmers sprayed strong poisons (pesticides).

Fewer frogs meant more insects, so farmers sprayed even more poison. Now many farmers protect frogs and use fewer pesticides, so the frogs can help them again.

Based on information about frogs and pest control in rice farming.

1 **Write the food chain from the story.** [1]

2 **What happened when there were fewer frogs?** [2]

3 **How can frogs help farmers?** [2]

REVIEW AND CHALLENGE · A5 · FOOD CHAINS

Show what you know

★★★
Grade 3

Name: Class: Date:

Learning goals

- Use food chain words.

1 Find the words.

S	P	K	S	W	B	A	P	F
C	O	N	S	U	M	E	R	S
R	L	I	O	W	E	U	I	U
S	V	V	C	O	Y	P	O	N
G	C	H	A	I	N	N	B	W
P	R	O	D	U	C	E	R	O
R	V	H	M	O	U	L	D	R
W	S	P	T	G	K	R	O	M
P	O	M	R	N	Y	Y	E	H

chain consumer mould producer sun worm

2 Every food chain starts with *a*... [1]

- ◯ plant
- ◯ tiger
- ◯ mushroom

3 Which is *a* decomposer? [1]

- ◯ goat
- ◯ mushroom
- ◯ grass

4 Challenge. Where does the energy in *a* food chain first come from? [1]

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
sort producers, consumers and decomposers	○	○	○
build a food chain	○	○	○
explain why decomposers matter	○	○	○
predict what happens when one animal disappears	○	○	○
test what makes bread mouldy	○	○	○